EXPLORATORY DATA ANALYSIS (EDA) REPORT

Used Car Price Intelligence & Prediction System

1. INTRODUCTION

Exploratory Data Analysis (EDA) is an important step in the data analysis process. It involves examining, visualizing and understanding a dataset to identify patterns, relationships, trends and unusual observations.

The objective of this analysis is to understand the factors that influence the resale prices of used cars in the Indian automobile market.

This analysis focuses on studying the distribution of car prices, understanding the relationship between selling prices and vehicle characteristics, and identifying patterns across different brands, fuel types, transmission types, ownership categories and locations.

The insights obtained from this analysis will help in understanding the used-car market and preparing the dataset for future machine learning applications.

2. OBJECTIVES OF EDA

The main objectives of this exploratory data analysis are:

1. To understand the distribution of used-car selling prices.
2. To analyze the relationship between manufacturing year and selling price.
3. To investigate how kilometers driven affect resale prices.
4. To compare prices across different car brands.
5. To study price variations based on fuel type.
6. To examine the relationship between transmission type and selling price.
7. To understand how the number of previous owners affects resale value.
8. To analyze price variations across different geographical locations.
9. To investigate the relationship between mileage, engine capacity, power and selling price.

3. DATASET OVERVIEW

10. The analysis uses the cleaned dataset generated in the previous data-cleaning stage.

Input Dataset: usedcars_stage2_cleaned.csv

The dataset contains information about used cars, including their names, locations, manufacturing years, kilometers driven, fuel types, transmission types, ownership details, mileage, engine capacity, power, seating capacity and selling prices.

The important attributes used in this analysis are:

- Name
- Location
- Manufacturing_Year
- Kilometers_Driven
- Fuel_Type
- Transmission
- Owner_Type
- Mileage
- Engine
- Power
- Seats
- Price

The selling price is the primary variable of interest in this project.

4. METHODOLOGY

The following Python libraries were used to perform exploratory data analysis:

- **Pandas:** Used for loading, manipulating and analyzing the dataset.
- **NumPy:** Used for numerical operations.
- **Matplotlib:** Used for creating graphs and visualizations.
- **Seaborn:** Used for statistical data visualization.
- **SciPy:** Imported for statistical operations.

The analysis was divided into two major categories:

1. Univariate Analysis
2. Bivariate Analysis

5. UNIVARIATE ANALYSIS

Univariate analysis examines individual variables to understand their distributions and characteristics.

5.1 Fuel Type Distribution

- **Visualization:** Count plot and pie chart.
- **Observation:** Diesel and petrol cars dominate the dataset, while other fuel types have fewer records.
- **Interpretation:** The dataset mainly represents petrol and diesel vehicles.

5.2 Owner Type Distribution

- **Visualization:** Count plot.
- **Observation:** First-owner cars are the most common, followed by cars with multiple previous owners.
- **Interpretation:** Most listings belong to first-owner vehicles.

5.3 Selling Price Distribution

- **Visualization:** Histogram with KDE.
- **Observation:** Price distribution is right-skewed, with most cars in lower price ranges and fewer expensive cars.
- **Interpretation:** Most listings are budget-friendly, while a few high-priced cars extend the price range.

5.4 Location Distribution

- **Visualization:** Pie chart.
- **Observation:** Mumbai and Hyderabad contribute more listings, while Jaipur and Bangalore contribute fewer.
- **Interpretation:** Car listings are distributed unevenly across different cities.

5.5 Manufacturing Year Distribution

- **Visualization:** Bar chart.
- **Observation:** Most listings belong to cars manufactured between 2013 and 2017.
- **Interpretation:** These manufacturing years are more strongly represented in the dataset.

6. BIVARIATE ANALYSIS

Bivariate analysis examines relationships between two variables to identify patterns and associations.

6.1 Manufacturing Year vs Selling Price

- **Visualization:** Regression plot.
- **Observation:** Newer cars generally have higher selling prices than older cars.
- **Interpretation:** Car age and depreciation are important factors influencing resale value.

6.2 Kilometers Driven vs Selling Price

- **Visualization:** Scatter plot.
- **Observation:** Selling prices generally decrease as kilometers driven increase.
- **Interpretation:** Higher vehicle usage may reduce resale value due to increased wear and tear.

6.3 Brand vs Selling Price

- **Visualization:** Box plot (top 12 brands).
- **Observation:** BMW, Audi and Mercedes-Benz occupy higher price ranges, while Tata, Maruti and Hyundai generally have lower price ranges.
- **Interpretation:** Brand influences resale value, although model and vehicle characteristics also contribute.

6.4 Fuel Type vs Selling Price

- **Visualization:** Box plot.
- **Observation:** Electric vehicles appear in higher price ranges, while petrol vehicles generally occupy lower ranges.
- **Interpretation:** Fuel type is associated with price variations, along with other vehicle characteristics.

6.5 Transmission Type vs Selling Price

- **Visualization:** Box plot.
- **Observation:** Automatic cars generally have higher selling prices than manual cars.
- **Interpretation:** Transmission type is associated with differences in resale value.

6.6 Owner Type vs Selling Price

- **Visualization:** Box plot.
- **Observation:** First-owner cars generally have higher prices, while prices tend to decrease with more previous owners. The Fourth & Above category shows a slight deviation.
- **Interpretation:** Ownership history influences resale value.

6.7 Location vs Selling Price

- **Visualization:** Bar chart (median selling price).
- **Observation:** Coimbatore, Bangalore and Kochi show higher median prices, while Pune, Jaipur and Kolkata show lower median prices.
- **Interpretation:** Location-wise differences may result from variations in demand and vehicle availability.

6.8 Mileage vs Selling Price

- **Visualization:** Regression plot.
- **Observation:** Mileage shows a negative relationship with selling price.
- **Interpretation:** Higher mileage values are generally associated with lower selling prices in the dataset.

6.9 Engine Capacity vs Selling Price

- **Visualization:** Regression plot.
- **Observation:** Engine capacity shows a positive relationship with selling price.
- **Interpretation:** Cars with larger engines generally tend to have higher selling prices.

6.10 Power vs Selling Price

- **Visualization:** Regression plot.
- **Observation:** Vehicle power shows a positive relationship with selling price.
- **Interpretation:** Higher-powered vehicles generally tend to have higher resale values.

7. KEY FINDINGS

Based on the exploratory data analysis, the following important findings were identified:

1. Diesel and petrol vehicles dominate the dataset.

2. First-owner vehicles represent the largest ownership category.

3. Used-car selling prices follow a right-skewed distribution.

4. Most listings are concentrated around manufacturing years 2013 to 2017.

5. Newer cars generally have higher selling prices than older cars.

6. Cars driven for more kilometers generally tend to have lower selling prices.

7. Premium brands such as BMW, Audi and Mercedes-Benz occupy higher price ranges.

8. Fuel type and transmission type are associated with variations in selling prices.

9. Vehicles with fewer previous owners generally have higher selling prices.

10. Median selling prices differ across geographical locations.

11. Mileage shows a negative relationship with selling price.

12. Engine capacity and vehicle power show positive relationships with selling price.

8. BUSINESS INSIGHTS

The EDA findings provide several useful insights for a used-car marketplace or dealership.

1. Vehicle Age: Manufacturing year is an important factor in estimating resale value. Older cars generally have lower selling prices.

2. Vehicle Usage: Kilometers driven can help buyers and sellers understand the usage history of a vehicle and its possible effect on resale value.

3. Brand Value: Different manufacturers occupy different price ranges. Brand information can help in estimating the expected market value.

4. Ownership History: Cars with fewer previous owners generally tend to retain higher resale values.

5. Vehicle Specifications: Fuel type, transmission, mileage, engine capacity and power are useful characteristics for understanding price variations.

6. Location: City-wise price differences indicate that geographical information can be useful when estimating the resale value of a vehicle.

These findings can be used to identify important input features for the future machine learning model.

9. CONCLUSION

The exploratory data analysis provided a better understanding of the used-car dataset and the different factors associated with resale prices.

The analysis identified important patterns involving manufacturing year, kilometers driven, brand, fuel type, transmission, ownership history, location, mileage, engine capacity and vehicle power.

The findings indicate that used-car prices are influenced by multiple vehicle characteristics rather than a single factor.

The visualizations helped identify relationships, distributions and variations within the dataset.

Overall, this EDA process provides a foundation for further statistical analysis and machine learning preprocessing in the next stages of the Used Car Price Intelligence & Prediction System.

10.TOOLS AND LIBRARIES USED

1. Python – Programming language.

2. Pandas – Data manipulation and analysis.

3. NumPy – Numerical computations.

4. Matplotlib – Data visualization.

5. Seaborn – Statistical visualization.

6. SciPy – Statistical operations.